

Two-round Walsh transforms as subspace embeddings

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Verification and AI disclosure. The embedding and moment bounds, their limited-independence forms, and the confidence obstruction have been formalized in Lean 4. The fully independent embedding interface has also passed Comparator and proof replay with NanoDa and Lean’s default kernel. Appendix B identifies the checked statements and the remaining Palomar rendering issue. The research, formalization, and manuscript were developed with substantial AI-agent assistance under the author’s direction. Independent human peer review is not claimed. No publication-priority or independent-discovery claim is made.

Abstract

Let H be the normalized Walsh–Hadamard matrix of order $n = 2^k$. We prove that a uniform sample of distinct rows from HD_yHD_x , with two independent diagonal sign matrices, embeds every fixed real d -dimensional subspace with squared-norm distortion ϵ and failure probability at most δ , using

$$M = \min \left\{ n, \lceil 8192\epsilon^{-2}(d + 27q^3) \rceil \right\}, \quad q = \left\lceil \log_4 \frac{4d}{\delta} \right\rceil.$$

The row count is $O(\epsilon^{-2}[d + \log^3(1/\delta)])$, and hence $O(d/\epsilon^2)$ at fixed failure probability. Each sign family needs only $\min(n, 4q)$ -wise independence. This establishes the constant-failure row order in Problem 5.6 of Amsel et al. [1] for the real Walsh model and the row-sampling convention specified here. Yang gives another proof of the constant-failure bound for the same two-round law, with a prescribed sample size [5, Theorem 1].

The proof represents the random Gram error by multiplication on a finite product space. A restricted-row normal-ordering identity and a weighted positive-operator inequality control this operator on the degrees reached by the relevant moment. A finite coupling and conditional Jensen inequality then transfer the Bernoulli trace-moment bound to sampling without replacement. We also give a support-alignment obstruction to a uniform $d + \log(1/\delta)$ row bound for any fixed number of Walsh/sign rounds. This obstruction does not establish that the cubic logarithm in our upper bound is optimal.

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1 Introduction

A subspace embedding compresses vectors while preserving their Euclidean geometry. If $U \in \mathbb{R}^{n \times d}$ has orthonormal columns, a random matrix $S \in \mathbb{R}^{M \times n}$ has the desired property when, with high probability,

$$(1 - \epsilon)\|z\|^2 \leq \|SUz\|^2 \leq (1 + \epsilon)\|z\|^2 \quad \text{for every } z \in \mathbb{R}^d.$$

The subspace is fixed before S is sampled. Equivalently, one must control the operator norm of the Gram error $U^\top S^\top S U - I_d$. This formulation controls the lower and upper spectral edges at once.

We study a transform obtained by alternating two normalized Walsh–Hadamard matrices with two independent diagonal sign matrices, and then retaining a uniform set of distinct rows:

$$S = \sqrt{\frac{n}{M}} R H D_y H D_x.$$

Here n is a power of two, and R is independent of both sign families. Before sampling, the transform is orthogonal for every realization of the signs. The question is how many rows suffice to preserve a prescribed d -dimensional subspace after sampling.

Our answer is $O(d/\epsilon^2)$ rows at fixed failure probability, uniformly over n and d . We give an explicit choice of the row count, track its dependence on the failure probability, and show that only bounded-coordinate sign moments are needed. The transform is the transpose of the real Walsh specialization of the rerandomized transform in Definition 5.3 of Amsel et al. [1]; Problem 5.6 asks for the constant-failure row order addressed here. Our theorem uses the normalization and sampling convention stated below. Complex Fourier transforms and other trigonometric systems require separate analysis.

Two endpoints help interpret the statement. If the displayed row bound reaches n , the theorem samples every row and the Gram matrix is exactly I_d ; this also covers $n = 1$. If δ is fixed and the bound is below n , the dependence on the subspace dimension is linear. The confidence dependence is a different question: the support alignment in Appendix A rules out a uniform linear-logarithmic confidence term for a fixed number of these rounds, while leaving a gap to the cubic logarithm in our upper bound.

Prior work and the scope of the comparison. For the traditional single-round SRHT, Tropp gives a spectral embedding bound and a rank-dependent logarithmic obstruction to injectivity [4, Theorem 3.1 and Section 3.3]. These concern one round, not the two-round law analyzed here. Alternating structured transforms and independent sign diagonals also predate this paper: they are used for angular locality-sensitive hashing by Andoni et al. [7] and in TripleSpin [8]. Those results concern different observables and do not, by themselves, prove the fixed-subspace theorem stated below.

Yang’s *Subspace embeddings with the rerandomized SRHT*, Theorem 1 [5], proves another constant-failure embedding bound for the same real Walsh law and uniform sampling without replacement. His prescribed width is $\min\{n, \lceil Cd/\epsilon^2 \rceil\}$ and his failure bound is 0.01, with the input subspace fixed before sampling. The transpose convention for the sketch does not change the subspace Gram matrix. The cited statement therefore overlaps the constant-failure conclusion here; it is not a result for a different sampling model.

Theorem 1.1 instead gives a stated conservative row choice for every $\delta \in (0, 1/2)$ and tracks sufficient independence orders for the sign families. These distinctions specify what is proved in this manuscript; they do not establish priority or the absence of other refinements. The TR-01 catalogue records the separate prescribed-width target and Yang’s reported verification evidence [6]. Neither that record nor this paper’s formal artifact is a certification of the other proof. No theorem of Yang or Tropp is used as a black box in the argument below.

1.1 Walsh normalization and the embedding event

Let $n = 2^k$, with $k \geq 0$, and index coordinates by the additive group $G = \mathbb{F}_2^k$. Addition of indices always means addition in this group. Write

$$\chi_b(a) = (-1)^{a \cdot b}, \quad H_{ab} = \frac{\chi_b(a)}{\sqrt{n}}, \quad f_b(a) = \frac{\chi_b(a)}{\sqrt{n}}. \quad (1.1)$$

The character identity $\chi_b(a)\chi_c(a) = \chi_{b+c}(a)$ and cancellation of every nontrivial character give

$$\sum_{a \in G} \chi_b(a)\chi_c(a) = n \mathbf{1}_{\{b=c\}}. \quad (1.2)$$

Indeed, if $b + c \neq 0$, choose a coordinate on which it is nonzero and pair each a with the vector obtained by flipping that coordinate. The paired summands cancel. Consequently $H^\top H = H^2 = I_n$, and $(f_b)_{b \in G}$ is an orthonormal basis.

Fix $U \in \mathbb{R}^{n \times d}$, where $1 \leq d \leq n$ and $U^\top U = I_d$. Define

$$u_i = U^\top e_i, \quad P = UU^\top, \quad \ell_i = \|u_i\|^2 = P_{ii}, \quad \mu = \max_i \ell_i. \quad (1.3)$$

Thus $P = P^\top = P^2$, $0 \leq \mu \leq 1$, $\sum_i \ell_i = d$, and $\sum_i u_i u_i^\top = I_d$. We use the uniform leverage bound $\ell_i \leq 1$; no improvement depending on the actual value of μ is required. Replacing μ by its upper bound one imposes no extra condition on the frame. All norms of operators or matrices below are Euclidean operator norms; tr is the usual unnormalized trace. The notation $A \preceq B$ means that $B - A$ is positive semidefinite.

Let $x = (x_i)$ and $y = (y_a)$ be sign families and put

$$W = HD_y HD_x U, \quad w_j = W^\top e_j, \quad \sum_j w_j w_j^\top = W^\top W = I_d. \quad (1.4)$$

The last equality holds for every realization of the signs. Let R select a uniform M -element subset of rows, in any fixed ordering of that subset, independently of both sign families. The sketch and its subspace Gram are

$$S = \sqrt{\frac{n}{M}} R H D_y H D_x, \quad U^\top S^\top S U = \frac{n}{M} \sum_{j \in T_M} w_j w_j^\top. \quad (1.5)$$

The event $\|U^\top S^\top S U - I_d\| \leq \epsilon$ is equivalent to

$$(1 - \epsilon)\|z\|^2 \leq \|S U z\|^2 \leq (1 + \epsilon)\|z\|^2 \quad \text{for every } z \in \mathbb{R}^d. \quad (1.6)$$

This is a statement for each fixed subspace chosen before the sketch randomness. It does not assert one sampled sketch works simultaneously for all subspaces.

1.2 Fixed-size theorem

Theorem 1.1 (Two-round Walsh embedding). *Let $n = 2^k$, $1 \leq d \leq n$, and $U^\top U = I_d$ as above. Fix $0 < \epsilon < 1$ and $0 < \delta < 1/2$, and define*

$$M = \min \left\{ n, \left\lceil 8192\epsilon^{-2}(d + 27q^3) \right\rceil \right\}, \quad q = \left\lceil \frac{\log(4d/\delta)}{2 \log 2} \right\rceil. \quad (1.7)$$

Put $D_q = d + 27q^3$. Suppose each sign family is unbiased and $\min(n, 4q)$ -wise independent, and the two families and the uniform row subset are mutually independent. Then

$$\mathbb{P} \left\{ \|U^\top S^\top S U - I_d\| > \epsilon \right\} \leq \delta. \quad (1.8)$$

There is no additional relation between n and d . When $M = n$, the Gram is exactly I_d ; in particular the case $n = 1$ is deterministic. The explicit constants are conservative proof constants.

Corollary 1.2 (Dimension and confidence dependence). *The row count in Theorem 1.1 satisfies*

$$M \leq \min \left\{ n, \frac{110592001}{\epsilon^2} [d + \log^3(1/\delta)] \right\}.$$

In particular, for every fixed $\delta \in (0, 1/2)$ it is $O(d/\epsilon^2)$, with a constant independent of n and d .

The proof, including integer rounding, is given in Section 11.2. The confidence dependence need not match that of an unstructured Gaussian embedding; Appendix A explains an obstruction specific to finitely many mixing rounds.

1.3 Bernoulli moment theorem

For $0 < \rho < 1$, let η_j be independent Bernoulli variables of mean ρ , independent of the signs. Set $m = \rho n$, an expected sample size that need not be an integer, and define

$$E_\rho = \frac{1}{\rho} \sum_j \eta_j w_j w_j^\top - I_d = \sum_j \left(\frac{\eta_j}{\rho} - 1 \right) w_j w_j^\top. \quad (1.9)$$

Theorem 1.3 (Bernoulli trace moments). *Under the fully independent product law, for every integer $q \geq 1$,*

$$\mathbb{E} \operatorname{tr}(E_\rho^{2q}) \leq d \left[6 \sqrt{\frac{d + 27q^3}{m}} + 9 \frac{d + 27q^3}{m} \right]^{2q}. \quad (1.10)$$

The same inequality holds when the two sign families are only $\min(n, 4q)$ -wise independent and the selector family is only $\min(n, 2q)$ -wise independent with the same Bernoulli marginals, provided the three families remain mutually independent. At $\rho = 1$, $E_\rho = 0$.

1.4 How the proof works

The main estimate concerns Bernoulli row sampling. Writing $w_j^\top$ for row j of $HD_y HD_x U$ and letting the selectors η_j have mean ρ , set

$$E_\rho = \sum_j (\eta_j / \rho - 1) w_j w_j^\top, \quad m = \rho n.$$

We bound its even trace moment by

$$\mathbb{E} \operatorname{tr}(E_\rho^{2q}) \leq d \left[6 \sqrt{\frac{d + 27q^3}{m}} + 9 \frac{d + 27q^3}{m} \right]^{2q}.$$

The important feature is the additive term $d + 27q^3$: the moment parameter does not multiply the dimension. The proof has four parts.

Finite-product representation. Under the independent product law, sign monomials form an orthonormal basis indexed by subsets of coordinates. Multiplication by a sign inserts or removes its coordinate from that subset. A centered Bernoulli selector has an equally explicit two-state multiplication matrix. Section 2 develops these identities and the associated extraction and insertion maps. This is finite-dimensional linear algebra; the occupation notation records polynomial degree.

Collective control of small row sets. Each row functional splits into four transitions according to whether it inserts or removes a coordinate from each sign family. Three transitions follow from the frame identity $U^\top U = I_d$. The double-annihilation transition must instead be collected over the row set before estimating its norm. Sections 3–5 derive the normal-ordering expansion, bound its five terms, and obtain control of every small row subset on a common sign-degree cutoff. This collective estimate retains the additive dependence on d .

Positive selector control and exact moments. The compressed row effects A_j are positive and sum to the identity. The weighted Cauchy–Schwarz argument of Section 6 converts small-subset control into a norm bound for selector creation without requiring the A_j to commute. Section 7 then identifies the trace moment with squared norms of q applications of the multiplication operator to constant functions. Such applications reach sign degrees at most $2q$ and selector degree at most q , so the cutoff preserves these vectors exactly. The moment estimate therefore concerns the original random matrix.

Sampling and probability. Section 8 transfers the moment by matching the coordinates used by each polynomial monomial. A separate finite coupling in Section 10 compares a Bernoulli subset with a uniform subset of exactly M rows. Conditional Jensen loses at most a factor of four in the $2q$ -th root of the trace moment. Section 11 applies Markov’s inequality, checks the endpoints and constants, and absorbs the apparent $\log^3 d$ term into the linear dimension term.

The sign families may thus have limited independence, but the final row set retains its full uniform law. The finite-product orthogonality is used only under the independent law; the extension to limited independence follows afterwards by exact moment matching.

1.5 Notation and conventions

All spaces are real and finite dimensional. Matrix and operator norms are Euclidean operator norms, and tr is the unnormalized trace. We write $A \preceq B$ when $B - A$ is positive semidefinite. Indices of Walsh coordinates belong to $G = \mathbb{F}_2^k$, and addition of indices is addition in this group. An adjoint is denoted by $\dagger$ in the occupation-space calculations and by $\top$ for coordinate matrices.

The letters r, s denote the grades of the two sign species. The selector grade counts the degree of a basis monomial, not the number of rows selected in a random realization. Both a deterministic row-set size and a selector grade are denoted by t locally; each use is specified where it occurs. The integer M is the actual sample size, whereas $m = \rho n$ is the expected size of a Bernoulli sample. These distinctions are useful throughout the operator and sampling arguments.

2 Finite-product Hilbert spaces and local operators

The moment argument begins by replacing random multiplication with an exact deterministic operator. The basis below records which coordinates occur in a monomial. Its grade is the polynomial degree, and the empty subset represents the constant function. We first treat independent coordinates; limited independence enters only after the moment has been expanded in Section 8.

Walsh–Fourier expansions on product spaces are classical [9]. The broader interpretation of moments as vacuum matrix elements is part of the interacting-Fock-space tradition [10]. Here these are elementary representation devices; the subsequent collected row and positive-selector estimates carry the application-specific bounds.

2.1 Sign monomials as an orthonormal occupation basis

Under independent unbiased signs, the monomials $x_A = \prod_{i \in A} x_i$, for $A \subseteq G$, form an orthonormal basis of the real L^2 space. In fact, $x_A x_B = x_{A \Delta B}$, whose expectation is zero unless $A = B$. Identify x_A with a basis vector $|A\rangle$. Let $\mathcal{X}_r$ be the span of subsets of size r ; define the second-sign spaces $\mathcal{Y}_s$ in the same way.

On a single occupation space define annihilation, creation, and occupancy by

$$\begin{aligned} a_i |A\rangle &= \mathbf{1}_{\{i \in A\}} |A \setminus \{i\}\rangle, \\ a_i^\dagger |A\rangle &= \mathbf{1}_{\{i \notin A\}} |A \cup \{i\}\rangle, \\ N_i |A\rangle &= \mathbf{1}_{\{i \in A\}} |A\rangle. \end{aligned} \tag{2.1}$$

Operators at different sites commute. Direct evaluation on the two local states gives

$$a_i^\dagger a_i = N_i, \quad a_i a_i^\dagger = I - N_i, \quad a_i a_l^\dagger = a_l^\dagger a_i + \mathbf{1}_{\{i=l\}} (I - 2N_i). \tag{2.2}$$

There are no fermionic signs. Multiplication by x_i toggles the presence of i , so its exact operator is $a_{x,i} + a_{x,i}^\dagger$. For a real vector $f \in \mathbb{R}^n$, write $a(f) = \sum_i f(i) a_i$, and similarly for creation. By bilinearity,

$$a(f) a(g)^\dagger = a(g)^\dagger a(f) + \langle f, g \rangle I - 2 \sum_i f(i) g(i) N_i. \tag{2.3}$$

In particular, for Walsh vectors,

$$a_y(f_b) a_y(f_c)^\dagger = a_y(f_c)^\dagger a_y(f_b) + \mathbf{1}_{\{b=c\}} I - \frac{2}{n} \sum_a \chi_{b+c}(a) N_{y,a}. \tag{2.4}$$

Example 2.1 (Multiplication changes occupation). For two sign coordinates, relabeled 1, 2, the orthonormal basis is $1, x_1, x_2, x_1 x_2$. Multiplication by x_1 sends

$$1 \longleftrightarrow x_1, \quad x_2 \longleftrightarrow x_1 x_2.$$

In subset notation these are the pairs $\emptyset \leftrightarrow \{1\}$ and $\{2\} \leftrightarrow \{1, 2\}$. Creation performs the transitions that insert coordinate 1, and annihilation performs those that remove it. Their sum is multiplication by x_1 because $x_1^2 = 1$.

2.2 Extraction, insertion, and their exact norm factors

For grade r , define an extraction map by

$$\mathcal{E}_r |A\rangle = \sum_{i \in A} e_i \otimes |A \setminus \{i\}\rangle : \mathcal{X}_r \longrightarrow \mathbb{R}^n \otimes \mathcal{X}_{r-1}. \tag{2.5}$$

Distinct input subsets have disjoint supports in the output basis, and each input produces r orthogonal terms. Thus $\mathcal{E}_r^\dagger \mathcal{E}_r = rI$ and $\|\mathcal{E}_r\| \leq \sqrt{r}$, with equality when the grade- r space is nonzero. The adjoint $\mathcal{E}_{r+1}^\dagger$ inserts a site, annihilating attempts to insert an already occupied site; its norm is at most $\sqrt{r+1}$, with equality when grade $r+1$ is nonzero. These upper bounds also cover empty grade spaces. The grade-zero extraction is zero. At the top grade, a creation map is zero.

2.3 One-particle operators and hard-core lifts

For a one-particle matrix $B = (B_{li})$, define its lift to grade r by

$$d\Gamma_r(B) = \sum_{l,i} B_{li} a_l^\dagger a_i. \tag{2.6}$$

To justify the bounds used later, identify $|A\rangle$, for $A = \{i_1, \dots, i_r\}$, with the normalized symmetric tensor

$$J_r|A\rangle = \frac{1}{\sqrt{r!}} \sum_{\pi \in \mathfrak{S}_r} e_{i_{\pi(1)}} \otimes \dots \otimes e_{i_{\pi(r)}}. \quad (2.7)$$

The vectors on the right are orthonormal as A varies. Compressing $\sum_{\nu=1}^r B^{(\nu)}$ by this isometry gives exactly (2.6): an off-diagonal entry replaces one occupied site by another absent site with coefficient B_{li} , while a diagonal entry contributes B_{ii} once for each occupied i . Repeated-site outputs are removed by compression.

It follows that

$$\|d\Gamma_r(B)\| \leq r\|B\|, \quad 0 \preceq B \preceq bI \implies 0 \preceq d\Gamma_r(B) \preceq rbI. \quad (2.8)$$

These assertions also hold for a one-particle operator acting jointly with an auxiliary row register: apply it to that register and one labeled particle at a time, sum over the particle slots, and compress. With two separate particle species of grades r, s , a joint row/one- x /one- y kernel K has a double lift of norm at most $rs\|K\|$. This follows by summing rs operators, one for each ordered pair of particle slots, before compression. For the separable matrix entries used below, the lift is precisely the corresponding product $a_{x,i}^\dagger a_{x,i} a_y(g)^\dagger a_y(f)$.

2.4 Biased selector sites

For a Bernoulli variable η of mean ρ , the local orthonormal basis is

$$|0\rangle = 1, \quad |1\rangle = \frac{\eta - \rho}{\sqrt{\rho(1 - \rho)}}. \quad (2.9)$$

Let $b = \eta/\rho - 1$. Its constant-to-constant matrix entry is zero. Its off-diagonal entry is $\mathbb{E}[b|1\rangle] = \sqrt{(1 - \rho)/\rho}$. Since

$$\mathbb{E}(\eta - \rho)^3 = \rho(1 - \rho)(1 - 2\rho), \quad (2.10)$$

the remaining diagonal entry is $(1 - 2\rho)/\rho$. Therefore multiplication by b is exactly

$$\begin{pmatrix} 0 & \sigma \\ \sigma & \tau \end{pmatrix} = \sigma(c^\dagger + c) + \tau N, \quad \sigma = \sqrt{\frac{1 - \rho}{\rho}}, \quad \tau = \frac{1 - 2\rho}{\rho}. \quad (2.11)$$

Here $c^\dagger, c, N$ are the same two-state creation, annihilation, and occupancy matrices as in (2.1). Products of the local basis vectors form an orthonormal basis for independent selectors; let $\mathcal{Z}_t$ denote selector degree t . Degree records a polynomial basis state, not the number of rows in an actual sampled realization.

Example 2.2 (An unbiased selector). When $\rho = 1/2$, the centered variable $\eta/\rho - 1 = 2\eta - 1$ is a uniform sign. The coefficients in the selector matrix become $\sigma = 1$ and $\tau = 0$, so multiplication is exactly $c + c^\dagger$. For a biased selector the occupancy term τN records the asymmetry of its two values. This is why selector control must include both degree-changing and degree-preserving terms.

3 Linear row factorization

The row functional contains one sign from each of the two families. It therefore splits into four occupation transitions. Three transitions are controlled by ordinary frame contractions. The remaining one removes one particle of each species and requires a collective estimate over the selected row labels; that is the purpose of Section 4.

3.1 Exact row functional and four grade transitions

We use the Walsh normalization and the isometric input frame from Section 1. In particular, $P = UU^\top$ satisfies $\|P\| \leq 1$, its diagonal entries obey $0 \leq \ell_i \leq 1$, and $\sum_i \ell_i = d$. All estimates in Sections 3–5 use this universal bound on the leverage scores.

Expanding both Walsh matrices gives the exact row identity

$$w_j^\top = \frac{1}{n} \sum_{a,i} \chi_j(a) \chi_i(a) y_a x_i u_i^\top = \frac{1}{\sqrt{n}} \sum_i u_i^\top X_i Y(f_{j+i}), \quad (3.1)$$

where X_i is multiplication by x_i , and $Y(f) = \sum_a f(a) y_a$ is multiplication by that linear form. Each row channel includes the common factor $n^{-1/2}$; Section 4 removes it explicitly when defining the unnormalized collected map. Under the occupation-basis identification in Section 2, $X_i = a_{x,i}^\dagger + a_{x,i}$ and $Y(f) = a_y(f)^\dagger + a_y(f)$.

For fixed input sign grades r, s , write

$$\mathcal{H}_{r,s} = \mathbb{R}^d \otimes \mathcal{X}_r \otimes \mathcal{Y}_s, \quad \Pi_{r,s} = \text{the orthogonal projection onto } \mathcal{H}_{r,s}. \quad (3.2)$$

Let $T_j^{\alpha\beta}$, for $\alpha, \beta \in \{+, -\}$, be the restriction of the corresponding term of (3.1) to $\mathcal{H}_{r,s}$, with $+$ denoting creation and $-$ annihilation. For example,

$$T_j^{--} = \frac{1}{\sqrt{n}} \sum_i u_i^\top a_{x,i} a_y(f_{j+i}) \Pi_{r,s}.$$

The four output grades are $(r+1, s+1)$, $(r+1, s-1)$, $(r-1, s+1)$, $(r-1, s-1)$. Any transition to an invalid grade is zero. Their output spaces are mutually orthogonal, so all cross-products between distinct channels vanish. Consequently

$$\Pi_{r,s} M_{w_j w_j^\top} \Pi_{r,s} = \sum_{\alpha, \beta \in \{+, -\}} (T_j^{\alpha\beta})^\dagger T_j^{\alpha\beta}. \quad (3.3)$$

Here M_F denotes multiplication by the matrix-valued function F . Thus (3.3) concerns the actual row multiplication operator, represented in the finite occupation basis.

3.2 Three coefficient flattenings

For a fixed row j , consider the coefficient tensor

$$C_j = \sum_i u_i^\top \otimes e_i \otimes f_{j+i}. \quad (3.4)$$

Its flattening $F_j : \mathbb{R}^d \rightarrow \mathbb{R}^n \otimes \mathbb{R}^n$ sends z to $\sum_i \langle u_i, z \rangle e_i \otimes f_{j+i}$. Orthogonality and $U^\top U = I_d$ give

$$\|F_j z\|^2 = \sum_i |\langle u_i, z \rangle|^2 = \|z\|^2. \quad (3.5)$$

For the flattening from the external and y factors to the x factor, express the input uniquely as $\sum_i z_i \otimes f_{j+i}$. The output is $\sum_i \langle u_i, z_i \rangle e_i$, and

$$\sum_i |\langle u_i, z_i \rangle|^2 \leq \sum_i \ell_i \|z_i\|^2 \leq \sum_i \|z_i\|^2. \quad (3.6)$$

The flattening from the external and x factors to the y factor is also a contraction: write the input in the basis (e_i) and use the orthonormality of (f_{j+i}) in the same calculation.

For the $++$ channel, apply F_j , then insert its two one-particle outputs into the sign occupations. The insertion norms from Section 2.2 are at most $\sqrt{r+1}$ and $\sqrt{s+1}$. For the $+-$ channel, extract one y site, apply the second flattening, and insert one x site. Its two

occupation factors are bounded by $\sqrt{s}$ and $\sqrt{r+1}$. Exchanging the two species gives the $-+$ channel. Restoring the common scalar in (3.1),

$$\begin{aligned}\|T_j^{++}\|^2 &\leq \frac{(r+1)(s+1)}{n}, \\ \|T_j^{+-}\|^2 &\leq \frac{(r+1)s}{n}, \\ \|T_j^{-+}\|^2 &\leq \frac{r(s+1)}{n}.\end{aligned}\tag{3.7}$$

These bounds remain valid at grade zero and at the top grade, because a forbidden extraction or insertion is the zero map.

3.3 Collecting the orthogonal channels

Set

$$a_{r,s} = (r+1)(s+1) + (r+1)s + r(s+1) = 3rs + 2r + 2s + 1.\tag{3.8}$$

Let $\mathcal{B}_j$ be the sum of the three channel Grams in (3.3) other than double annihilation. Each Gram is positive, and (3.7) gives

$$0 \preceq \mathcal{B}_j \preceq \frac{a_{r,s}}{n} I.\tag{3.9}$$

For a deterministic set of row labels $T \subseteq G$, with $t = |T|$, define

$$K_T = \sum_{j \in T} M_{w_j w_j^\top}, \quad \mathcal{R}_T z = \sum_{j \in T} e_j \otimes T_j^{--} z.\tag{3.10}$$

The identity $W^\top W = I_d$, which holds for every sign realization, implies $0 \preceq K_T \preceq I$. Summing the exact orthogonal-channel identity yields

$$\begin{aligned}\Pi_{r,s} K_T \Pi_{r,s} &= \mathcal{R}_T^\dagger \mathcal{R}_T + \sum_{j \in T} \mathcal{B}_j, \\ \|\Pi_{r,s} K_T \Pi_{r,s}\| &\leq \|\mathcal{R}_T\|^2 + \frac{ta_{r,s}}{n}.\end{aligned}\tag{3.11}$$

Equivalently, one can first collect each of the four channels over T , use their orthogonal output grades to eliminate all cross-products, and then sum the four Gram norms. Both expressions give (3.11). It remains to control the double-annihilation channel collectively over the row set.

4 Restricted double annihilation

The point of collecting the row labels before taking a norm is to pay the dimension term only once. The estimate below is the main use of the Walsh structure beyond orthogonality.

Proposition 4.1 (Restricted double annihilation). *Let $T \subseteq G$, $t = |T|$, and let $\mathcal{R}_T$ be the collected double-annihilation map in (3.10). On sign grades $r, s \geq 1$, write $p = r - 1$ and $v = s - 1$. Then*

$$\|\mathcal{R}_T\|^2 \leq \frac{d + 3p + pv + tv}{n}.$$

If either sign grade is zero, the map is zero.

We prove the proposition by expanding the collected Gram into five terms. The correction carrying a minus sign is estimated in norm along with the other terms.

4.1 Collected Gram and normal-ordering expansion

Assume first that $r, s \geq 1$, and put $p = r - 1$ and $v = s - 1$. These are the output grades of double annihilation; $t = |T|$ continues to denote the number of selected row labels. Define the unnormalized collected map on the full sign spaces by

$$(\mathcal{A}_T z)_j = \sum_i u_i^\top a_{x,i} a_y(f_{j+i}) z, \quad j \in T.$$

Thus $\mathcal{R}_T = n^{-1/2} \mathcal{A}_T \Pi_{r,s}$. Its uncompressed row Gram $\mathcal{G}_T = \mathcal{A}_T \mathcal{A}_T^\dagger$ has blocks

$$(\mathcal{G}_T)_{jk} = \sum_{i,l} P_{il} a_{x,i} a_{x,l}^\dagger a_y(f_{j+i}) a_y(f_{k+l})^\dagger, \quad j, k \in T. \quad (4.1)$$

Let $Q_{p,v}$ be the projection onto $\mathbb{R}^T \otimes \mathcal{X}_p \otimes \mathcal{Y}_v$. Double annihilation has the exact grade transition $(r, s) \mapsto (p, v)$. Therefore

$$\mathcal{A}_T \Pi_{r,s} = Q_{p,v} \mathcal{A}_T \Pi_{r,s}, \quad n \|\mathcal{R}_T\|^2 \leq \|Q_{p,v} \mathcal{A}_T\|^2 = \|Q_{p,v} \mathcal{G}_T Q_{p,v}\|.$$

This compression inequality is sufficient for the proof and keeps the input and output projections explicit.

Introduce $S_i = I - 2N_{x,i}$ and

$$\Delta_{bc} = \mathbf{1}_{\{b=c\}} I - \frac{2}{n} \sum_a \chi_{b+c}(a) N_{y,a}. \quad (4.2)$$

The commutation identities (2.2) and (2.4) expand each row block of (4.1) as

$$\sum_{i,l} P_{il} (a_{x,l}^\dagger a_{x,i} + \mathbf{1}_{\{i=l\}} S_i) (a_y(f_{k+l})^\dagger a_y(f_{j+i}) + \Delta_{j+i,k+l}). \quad (4.3)$$

The product of the two normally ordered factors is $\mathcal{M}$; the x commutator times the normally ordered y factor is $\mathcal{Y}$; and the normally ordered x factor times Δ splits into $\mathcal{X} - \mathcal{F}$. The product of the commutators is $\mathcal{E}_0$. In the next subsections, these symbols denote the corresponding operators restricted to the output grades p, v . Every term preserves those two grades, and the exact identity is

$$Q_{p,v} \mathcal{G}_T Q_{p,v} = \mathcal{E}_0 + \mathcal{X} - \mathcal{F} + \mathcal{Y} + \mathcal{M}. \quad (4.4)$$

We bound the norm of each of these five terms, including the correction $\mathcal{F}$ that appears with a minus sign.

4.2 The grouped double commutator

Let $L_x = \sum_i \ell_i N_{x,i}$, so $0 \preceq L_x \preceq dI$. Since $i = l$ in the product of the two commutators, $\chi_{j+i+k+i}(a) = \chi_{j+k}(a)$, and its row blocks are

$$(\mathcal{E}_0)_{jk} = (dI - 2L_x) \left[\mathbf{1}_{\{j=k\}} I - \frac{2}{n} \sum_a \chi_{j+k}(a) N_{y,a} \right]. \quad (4.5)$$

On the full row register define

$$Q = \sum_a |f_a\rangle \langle f_a|_{\text{row}} \otimes N_{y,a}. \quad (4.6)$$

The Walsh vectors are orthonormal and the occupancy operators are projections, so $Q^2 = Q = Q^\dagger$. Its compression Q_T to row labels in T satisfies $0 \preceq Q_T \preceq I$. Its row block j, k is $n^{-1} \sum_a \chi_{j+k}(a) N_{y,a}$. Hence

$$\mathcal{E}_0 = (dI - 2L_x)(I - 2Q_T), \quad \|\mathcal{E}_0\| \leq d. \quad (4.7)$$

The factors act on separate x and row/ y registers and commute. Their spectral intervals lie in $[-d, d]$ and $[-1, 1]$. Equivalently, on a fixed pair of occupation subsets the first factor is the scalar $\sum_i \ell_i (1 - 2\mathbf{1}_{\{i \in A\}})$, whose absolute value is at most d , while the second factor is a compression of a Walsh-conjugated diagonal sign matrix. Thus grade compression introduces no additional norm factor.

4.3 The x -transfer term

The identity part of the y commutator contributes

$$\mathcal{X}_{jk} = \sum_{i,l} P_{il} \mathbf{1}_{\{j+i=k+l\}} a_{x,l}^\dagger a_{x,i}. \quad (4.8)$$

Its row/one-particle kernel is

$$\mathbf{X}_{jk} = \sum_{i,l} P_{il} \mathbf{1}_{\{j+i=k+l\}} |l\rangle \langle i|. \quad (4.9)$$

First allow all row labels in G . Fourier transformation of the row register gives

$$\frac{1}{n} \sum_{j,k} \chi_a(j) \chi_b(k) \mathbf{1}_{\{j+i=k+l\}} = \mathbf{1}_{\{a=b\}} \chi_a(i+l). \quad (4.10)$$

Indeed, substitute $k = j + i + l$ and apply character orthogonality. With $D_a = \text{diag}(\chi_a(i))$, the frequency- a block is $D_a P D_a$, of norm at most one. Restriction to T is an orthogonal compression, so the restricted kernel is also a contraction. Its lift to the p occupied x slots has norm at most p by Section 2.3:

$$\|\mathbf{X}\| \leq 1, \quad \|\mathcal{X}\| \leq p. \quad (4.11)$$

The second sign register is a spectator in this estimate.

4.4 Bounding the correction term in norm

The remaining part of the y commutator gives the correction

$$\mathcal{F} = \frac{2}{n} \sum_a [\chi_{j+k}(a)]_{j,k \in T} \otimes d\Gamma_p(D_a P D_a) \otimes N_{y,a}. \quad (4.12)$$

This formula follows by separating $\chi_{j+i+k+l}(a)$ into $\chi_{j+k}(a) \chi_i(a) \chi_l(a)$. To estimate its norm, first replace the lifted operator in (4.12) by the one-particle matrix $D_a P D_a$, and allow all row labels. The resulting kernel on the row, one- x -particle, and full y -occupation registers is

$$\mathbf{F}_{\text{full}} = 2 \sum_a |f_a\rangle \langle f_a|_{\text{row}} \otimes D_a P D_a \otimes N_{y,a}.$$

Fourier transformation of the row register makes this operator a direct sum of the blocks $2D_a P D_a \otimes N_{y,a}$. Each block has norm at most two, because $\|P\| \leq 1$ and $\|N_{y,a}\| \leq 1$. Row restriction cannot increase this norm. Applying the single-particle lift on grade p , with the row and y -occupation registers retained as auxiliary factors, gives

$$\|\mathbf{F}\| \leq 2, \quad \boxed{\|\mathcal{F}\| \leq 2p}. \quad (4.13)$$

In the norm estimate for (4.4), the term $-\mathcal{F}$ therefore contributes $2p$. Its sign does not remove the need to control its norm.

4.5 The y -transfer term

The next term is

$$\mathcal{Y}_{jk} = \sum_i \ell_i S_i \otimes a_y(f_{k+i})^\dagger a_y(f_{j+i}). \quad (4.14)$$

Each $S_i = I - 2N_{x,i}$ is a self-adjoint unitary and commutes with the y operators. For $\psi = (\psi_j)_{j \in T}$ in the output-grade space, moving one annihilator to each side gives

$$\langle \psi, \mathcal{Y}\psi \rangle = \sum_{i,j,k} \ell_i \langle a_y(f_{k+i})\psi_j, S_i a_y(f_{j+i})\psi_k \rangle. \quad (4.15)$$

Apply Cauchy–Schwarz to the complete sum over (i, j, k) , with weights ℓ_i . The two sums of squared norms agree after exchanging j, k , since S_i is unitary. Thus

$$|\langle \psi, \mathcal{Y}\psi \rangle| \leq \sum_j \langle \psi_j, d\Gamma_v(G_T)\psi_j \rangle, \quad G_T = \sum_i \ell_i \sum_{k \in T} |f_{k+i}\rangle \langle f_{k+i}|. \quad (4.16)$$

For each fixed k , the vectors f_{k+i} form an orthonormal basis as i varies. Since $\ell_i \leq 1$, the corresponding weighted sum of rank-one projections is at most I . It follows that $0 \leq G_T \leq tI$. The lift on grade v has norm at most tv . Also, $\mathcal{Y}$ is self-adjoint, as is seen by transposing its row blocks and exchanging j, k . Hence

$$\|\mathcal{Y}\| \leq tv. \quad (4.17)$$

The full commutator S_i , of norm one, is retained throughout this estimate.

4.6 The mixed kernel is a contraction

The fully normally ordered term is

$$\mathcal{M}_{jk} = \sum_{i,l} P_{il} a_{x,l}^\dagger a_{x,i} a_y(f_{k+l})^\dagger a_y(f_{j+i}). \quad (4.18)$$

Its row/one- x /one- y kernel is

$$\mathbf{K}_{jk} = \sum_{i,l} P_{il} |l\rangle \langle i|_x \otimes |f_{k+l}\rangle \langle f_{j+i}|_y. \quad (4.19)$$

The row labels in this expression remain restricted to T . To factor the kernel, write an input as $(z_{k,i})_{k \in T, i \in G}$, with each $z_{k,i} \in \mathbb{R}_y^n$. First perform Walsh analysis in the y factor:

$$z'_{k,i,j} = \langle f_{j+i}, z_{k,i} \rangle, \quad j \in T. \quad (4.20)$$

For each k, i , this is analysis against an orthonormal family, so its norm is at most one. Next apply P to the one- x -particle index and exchange the two row labels:

$$z''_{j,l,k} = \sum_i P_{li} z'_{k,i,j}. \quad (4.21)$$

For each pair k, j , this is a contraction because $\|P\| \leq 1$; exchanging the row registers is an isometry. Finally synthesize in the y factor:

$$z'''_{j,l} = \sum_{k \in T} f_{k+l} z''_{j,l,k}. \quad (4.22)$$

For each j, l , synthesis in this orthonormal family is an isometry. Composing (4.20)–(4.22) gives precisely (4.19), using $P_{li} = P_{il}$. Thus $\|\mathbf{K}\| \leq 1$.

Extracting one particle from each sign occupation has norm at most $\sqrt{pv}$. Apply K to the row and extracted-particle registers while leaving the remaining subsets unchanged, and then apply the adjoint extraction map. The resulting operator has exactly the entries in (4.18). This also follows from the double-lift description in Section 2.3. Therefore

$$\|\mathcal{M}\| \leq pv\|K\| \leq pv. \quad (4.23)$$

The shared row register is part of the contraction, so this step introduces no additional factor of t or d .

4.7 The restricted-row estimate

Apply the triangle inequality to the exact five-term identity (4.4). Equations (4.7), (4.11), (4.13), (4.17), and (4.23) give

$$\|Q_{p,v}\mathcal{G}_T Q_{p,v}\| \leq d + p + 2p + tv + pv = d + 3p + pv + tv. \quad (4.24)$$

Combining this with the compression inequality following (4.1) proves

$$\boxed{\begin{aligned} \|\mathcal{R}_T\|^2 &\leq \frac{d + 3p + pv + tv}{n} \\ &= \frac{d + 3(r-1) + (r-1)(s-1) + t(s-1)}{n}, \quad r, s \geq 1. \end{aligned}} \quad (4.25)$$

If $r = 0$ or $s = 0$, double annihilation is zero. If T is empty, the collected map is also zero. For a uniform expression valid for all grades, the right side of (4.25) can use $p = \max(r-1, 0)$ and $v = \max(s-1, 0)$; it is nonnegative and therefore also bounds the zero-map cases. The estimate holds for every deterministic row-label subset T .

5 Small row subsets on a common sign cutoff

The selector estimate will need control of several row effects on one common sign space. A bound on an individual diagonal grade block is not enough: different grades can interact. Positivity and the fact that each grade has at most nine neighbors provide the required passage.

Proposition 5.1 (Small subsets on the sign cutoff). *Let $q \geq 1$, $D_q = d + 27q^3$, and let P_{sig} retain both sign grades at most $\min(2q, n)$. For every $T \subseteq G$ with $|T| \leq \min(q, n)$,*

$$\|P_{\text{sig}} K_T P_{\text{sig}}\| \leq \frac{9D_q}{n}.$$

The proof first bounds the diagonal grade blocks and then combines them.

5.1 Diagonal sign-grade bound

For all $r, s \geq 0$, put $p = \max(r-1, 0)$ and $v = \max(s-1, 0)$. Combining the orthogonal-channel decomposition (3.11) with (4.25) and its zero-grade cases gives

$$\|\Pi_{r,s} K_T \Pi_{r,s}\| \leq \frac{d + 3p + pv + tv + ta_{r,s}}{n}. \quad (5.1)$$

When a sign grade is zero, the double-annihilation contribution is actually zero; the displayed uniform upper bound remains valid. Each bound may also be capped at one because $0 \preceq K_T \preceq I$.

Fix an integer $q \geq 1$, and suppose $r, s \leq 2q$ and $t \leq q$. Then $p, v \leq 2q - 1$, and every factor in the occupation terms of (5.1) is nonnegative. Using (3.8),

$$\begin{aligned} a_{r,s} &\leq 12q^2 + 8q + 1, \\ 3p + pv + tv + ta_{r,s} &\leq 3(2q - 1) + (2q - 1)^2 + q(2q - 1) + q(12q^2 + 8q + 1) \\ &= 12q^3 + 14q^2 + 2q - 2. \end{aligned} \quad (5.2)$$

The exact slack in the desired cubic allowance is

$$27q^3 - (12q^3 + 14q^2 + 2q - 2) = 15q^2(q - 1) + (q - 1)^2 + 1 \geq 0.$$

The correction term is therefore covered by the cubic allowance, and

$$\|\Pi_{r,s} K_T \Pi_{r,s}\| \leq \frac{d + 12q^3 + 14q^2 + 2q - 2}{n} \leq \frac{d + 27q^3}{n} = \frac{D_q}{n}. \quad (5.3)$$

This includes all zero-grade and empty-row-set cases.

5.2 Positive block-Schur bound

This is a finite positive-block version of the classical Schur-test principle [11]. The proof keeps the grade adjacency and the positive quadratic forms explicit; the general norm-control principle is not claimed as new. Let P_{sig} keep both sign grades at most $\min(2q, n)$, acting as the identity on the external factor, and put $A = P_{\text{sig}} K_T P_{\text{sig}} \succeq 0$ and $b = D_q/n$. For every grade pair $\alpha = (r, s) \in \{0, \dots, n\}^2$, let $P_\alpha = \Pi_{r,s}$. These projections are mutually orthogonal and sum to the identity on the full sign space, including the external factor. They commute with P_{sig} . Thus (5.3) gives $\|P_\alpha A P_\alpha\| \leq b$ on every retained grade; on an excluded grade this block is zero.

Write $\alpha \sim \beta$ when each coordinate of the two grade pairs is equal or differs by exactly two. Each row functional changes each sign occupation by exactly $+1$ or -1 . A nonzero Gram entry therefore connects two inputs with a common output grade, forcing their occupation sizes to differ by 0 or ± 2 in each family. Consequently, $P_\alpha A P_\beta = 0$ unless $\alpha \sim \beta$. This relation is symmetric and each grade has at most nine neighbors, including itself. Grades outside $0, \dots, n$ are absent, so boundary grades can only have fewer neighbors.

For an arbitrary vector z , set $z_\alpha = P_\alpha z$. Then $\sum_\alpha z_\alpha = z$ and $\sum_\alpha \|z_\alpha\|^2 = \|z\|^2$. Positivity applied to $z_\alpha - z_\beta$, followed by the diagonal block bounds, gives

$$\begin{aligned} 2\langle z_\alpha, A z_\beta \rangle &\leq \langle z_\alpha, A z_\alpha \rangle + \langle z_\beta, A z_\beta \rangle \\ &\leq b(\|z_\alpha\|^2 + \|z_\beta\|^2). \end{aligned} \quad (5.4)$$

Sum over ordered neighboring pairs. Symmetry counts each grade's squared norm once for each neighbor in each of the two positions. Writing $\deg(\alpha) = |\{\beta : \alpha \sim \beta\}|$, we obtain

$$\begin{aligned} 2\langle z, A z \rangle &= 2 \sum_{\alpha \sim \beta} \langle z_\alpha, A z_\beta \rangle \\ &\leq b \sum_{\alpha \sim \beta} (\|z_\alpha\|^2 + \|z_\beta\|^2) \\ &= 2b \sum_{\alpha} \deg(\alpha) \|z_\alpha\|^2 \\ &\leq 18b \|z\|^2. \end{aligned} \quad (5.5)$$

Hence $\langle z, A z \rangle \leq 9b \|z\|^2$.

For completeness, a positive operator whose quadratic form is bounded by $C\|z\|^2$, with $C \geq 0$, has norm at most C . Indeed, writing $Q(z) = \langle z, A z \rangle$, positivity and polarization give, for unit vectors x, y ,

$$4\langle x, A y \rangle = Q(x + y) - Q(x - y) \leq C\|x + y\|^2 \leq 4C.$$

Replacing x by $-x$ bounds the absolute value, and taking the supremum proves the norm bound. The zero-dimensional case is immediate. Applying this with $C = 9b$ proves

$$\|P_{\text{sig}} K_T P_{\text{sig}}\| \leq L, \quad L = \frac{9D_q}{n}, \quad |T| \leq \min(q, n). \quad (5.6)$$

This proves Proposition 5.1. Its factor nine is the maximum number of neighboring grade pairs and is independent of the total number of grades. It gives the deterministic small-subset estimate used in Section 6, with the same $D_q = d + 27q^3$ and the same subsequent constants.

6 A positive-operator selector lemma

We now separate the sampling argument from the Walsh calculation. The input to this section is a positive family whose total mass is at most the identity and whose small subsets have controlled norm. Keeping these positive weights inside Cauchy–Schwarz is what avoids a factor proportional to the number of selectors.

6.1 Abstract statement

Lemma 6.1 (Positive selector control). *Let $A_j \succeq 0$ be operators on a finite-dimensional Hilbert space $\mathcal{H}$, with $\sum_j A_j = I_{\mathcal{H}}$. Suppose*

$$\left\| \sum_{j \in T} A_j \right\| \leq L \quad \text{whenever } |T| \leq \min(q, n). \quad (6.1)$$

Compress the selector occupation space to degrees at most $\min(q, n)$. On this space define $C = \sum_j A_j \otimes c_j^\dagger$ and $N_A = \sum_j A_j \otimes N_j$, with compression on both sides. Then

$$\|C\| \leq \sqrt{L}, \quad \|C^\dagger\| \leq \sqrt{L}, \quad \|N_A\| \leq L. \quad (6.2)$$

The A_j need not commute. The same conclusion holds if their sum is merely bounded above by the identity.

6.2 Weighted Cauchy-Schwarz proof

Proof. Restrict creation from selector grade t to grade $t + 1$, where $t + 1 \leq \min(q, n)$. Write an input as $(h_S)_{|S|=t}$. At output subset T ,

$$(Ch)_T = \sum_{j \in T} A_j h_{T \setminus \{j\}}. \quad (6.3)$$

The row operator $(v_j)_{j \in T} \mapsto \sum_{j \in T} A_j^{1/2} v_j$ has squared norm $\|\sum_{j \in T} A_j\|$. Apply it to $v_j = A_j^{1/2} h_{T \setminus \{j\}}$ to obtain

$$\|(Ch)_T\|^2 \leq L \sum_{j \in T} \langle h_{T \setminus \{j\}}, A_j h_{T \setminus \{j\}} \rangle. \quad (6.4)$$

Summing over T and writing $S = T \setminus \{j\}$ yields

$$\begin{aligned} \|Ch\|^2 &\leq L \sum_{|S|=t} \left\langle h_S, \left(\sum_{j \notin S} A_j \right) h_S \right\rangle \\ &\leq L \sum_{|S|=t} \|h_S\|^2. \end{aligned} \quad (6.5)$$

Outputs from distinct input grades have distinct grades, so their squared norms add. Creation from the top permitted grade is discarded. Taking the compressed adjoint proves the annihilation estimate. On each selector subset T , N_A acts as $\sum_{j \in T} A_j$, proving its bound.

The positive weights A_j in (6.4) are what allow their sum to be bounded by the identity in (6.5). Replacing them by scalar norm bounds before summing would lose this information. Only subsets of size at most q are used, because every surviving creation output lies below that cutoff. $\square$

Example 6.2 (Equal positive weights). Take $A_j = I_{\mathcal{H}}/n$ for every j . On selector degrees at most $q \leq n$, every relevant row subset has sum at most $(q/n)I_{\mathcal{H}}$, so the lemma gives $\|C\| \leq \sqrt{q/n}$ and $\|N_A\| \leq q/n$. The creation bound uses the normalization $\sum_j A_j = I_{\mathcal{H}}$ together with the subset size. The proof keeps this mechanism when the positive weights are unequal and noncommuting.

6.3 Centered Bernoulli consequence

Using the exact local matrix (2.11) and the triangle inequality, Lemma 6.1 gives

$$\left\| \sum_j A_j \otimes [\sigma(c_j^\dagger + c_j) + \tau N_j] \right\| \leq 2\sigma\sqrt{L} + |\tau|L. \quad (6.6)$$

This simultaneously controls all selector transitions.

7 Assembly of the full moment bound

The preceding two sections fit together because compression preserves positivity and the row effects still sum to the identity. We first bound the compressed multiplication operator and then show that it gives the exact moment of the original Gram error.

7.1 Positive row effects after sign compression

On $\mathcal{H} = \text{range}(P_{\text{sig}})$, set

$$A_j = P_{\text{sig}} M_{w_j w_j^\top} P_{\text{sig}}. \quad (7.1)$$

Every A_j is positive, since it is a compression of a positive multiplication operator. Pointwise orthogonality (1.4) gives $\sum_j A_j = I_{\mathcal{H}}$. Equation (5.6) gives the subset hypothesis of Lemma 6.1 with $L = 9D_q/n$.

Let P_{sel} keep selector grades at most $\min(q, n)$, and put $P_{\text{cut}} = P_{\text{sig}} \otimes P_{\text{sel}}$. In the exact product basis, let $\mathcal{M}$ denote multiplication by the matrix E_ρ . Its compression $\mathcal{M}_{\text{cut}} = P_{\text{cut}} \mathcal{M} P_{\text{cut}}$, restricted to the range, is precisely the operator in (6.6). Hence

$$\begin{aligned} \|\mathcal{M}_{\text{cut}}\| &\leq 2\sqrt{\frac{1-\rho}{\rho}} \sqrt{\frac{9D_q}{n}} + \frac{|1-2\rho|}{\rho} \frac{9D_q}{n} \\ &\leq 6\sqrt{\frac{D_q}{m}} + 9\frac{D_q}{m}. \end{aligned} \quad (7.2)$$

Both sign families remain inside the positive compressed row effects. No supremum over sign realizations is taken.

7.2 Exact vacuum identity and why the cutoff is sufficient

Let Ω be the constant function in all sign and selector coordinates. Under the product-law inner product, the map that represents functions in the orthonormal occupation basis is an isometry. Since E_ρ is symmetric pointwise,

$$\begin{aligned}\mathbb{E} \operatorname{tr}(E_\rho^{2q}) &= \sum_{a=1}^d \mathbb{E} \|E_\rho^q e_a\|^2 \\ &= \sum_{a=1}^d \|\mathcal{M}^q(e_a \otimes \Omega)\|^2.\end{aligned}\tag{7.3}$$

One application of $\mathcal{M}$ changes each sign degree by at most two and selector degree by at most one. Starting from the vacuum, after j applications the sign degrees are at most $2j$ and the selector degree at most j . Therefore every intermediate vector for $0 \leq j \leq q$ lies in $\operatorname{range}(P_{\text{cut}})$. Induction gives the exact vector identity

$$\mathcal{M}_{\text{cut}}^q(e_a \otimes \Omega) = \mathcal{M}^q(e_a \otimes \Omega).\tag{7.4}$$

Only q applications are followed; the even moment is then obtained by squaring a norm. We do not need to preserve arbitrary length- $2q$ paths after an intermediate cutoff, nor assert any bound on the norm of the full uncompressed multiplication operator.

Combining (7.2)-(7.4), with $\|e_a \otimes \Omega\| = 1$, proves

$$\mathbb{E} \operatorname{tr}(E_\rho^{2q}) \leq d \|\mathcal{M}_{\text{cut}}\|^{2q} \leq d \left[6\sqrt{D_q/m} + 9D_q/m \right]^{2q}.\tag{7.5}$$

This proves Theorem 1.3 under full independence.

8 Exact transfer to limited independence

Only a finite set of coordinate moments is used in an even trace moment. This is the elementary low-degree moment-matching principle for product coordinates; see the Fourier and pseudorandomness treatment in [9]. It transfers the bound without constructing an occupation basis for the limited-independence law. The content specific to this application is the separate coordinate count in the three families, not the moment-matching identity itself. Each coordinate of w_j has degree one in x and degree one in y . Thus each entry of E_ρ has degree at most two in each sign family and at most one in the selectors. After expanding a product of $2q$ such entries, each monomial contains at most $4q$ distinct coordinates from either sign family and at most $2q$ distinct selector coordinates.

Repeated coordinates do not increase those counts: $x_i^2 = y_i^2 = 1$ and $\eta_j^a = \eta_j$ for every integer $a \geq 1$. Consequently $\operatorname{tr}(E_\rho^{2q})$ is a multilinear polynomial of separate degrees at most $4q, 4q, 2q$.

For a family that is k -wise independent with the specified marginals, every monomial involving at most k distinct family members has the same expectation as under full independence. Mutual independence of the three families then makes each cross-family monomial expectation factor in the same way. Linearity of expectation gives exact moment equality with the product law.

This transfers (7.5) to the distributions in Theorem 1.3. The cutoffs and operator estimates were proved only under the product law; no orthogonality of high-degree monomials under a limited-independence distribution is assumed.

For Theorem 1.1, only the signs are replaced by limited-independence families. The auxiliary Bernoulli subset in the finite sampling coupling has fully independent selectors. The fixed-size trace moment also has separate sign degrees at most $4q, 4q$, so it matches the fully independent

sign moment with the same uniform row law. The comparison in Section 10 can therefore be transferred to the required sign laws. This argument does not derandomize the final uniform subset of distinct rows.

9 A finite Bernoulli tail bound

We record the direct Bernoulli consequence before passing to a sample of fixed size. The same spectral argument will finish the main theorem after the sampling comparison. Let $0 < a < 1$. If $m \geq 256D_q/a^2$, then

$$6\sqrt{D_q/m} + 9D_q/m \leq \frac{3}{8}a + \frac{9}{256}a^2 \leq \frac{105}{256}a < \frac{a}{2}. \quad (9.1)$$

For every real symmetric matrix B , an eigenvalue of absolute value larger than a forces $\text{tr}(B^{2q}) > a^{2q}$, because all even eigenvalue powers are nonnegative. Markov's inequality and Theorem 1.3 therefore give

$$\mathbb{P}\{\|E_\rho\| > a\} \leq \frac{\mathbb{E} \text{tr}(E_\rho^{2q})}{a^{2q}} \leq d 2^{-2q}. \quad (9.2)$$

For the choice of q in Theorem 1.1, $d 2^{-2q} \leq \delta/4$.

10 Fixed-size sampling without replacement

Bernoulli sampling has independent selectors, whereas the sketch in Theorem 1.1 uses exactly M distinct rows. We connect these two laws through a finite coupling. Its conditional mean retains at least one quarter of the fixed-size Gram error, so convexity of an even trace power gives a comparison with an absolute constant.

There is classical precedent for coupling finite-population samples and using convexity to compare their moments: Hoeffding's Theorem 4 [12] compares sampling with and without replacement, and Gross–Nesme make its matrix application explicit [13]. The comparison here is different: it couples a Bernoulli subset to a uniform subset of prescribed size and charges the computed retention factor. It is proved below rather than identified with a factor-one consequence of the with-replacement comparison.

10.1 A finite nested fill/thin coupling

Fix an integer $1 \leq M < n$ and put $\rho = M/n$. In this section the row vectors w_j are fixed and satisfy $\sum_{j \in G} w_j w_j^\top = I_d$. Write

$$\begin{aligned} A_j &= w_j w_j^\top, & \sum_{j \in G} A_j &= I_d, \\ E_\rho(B) &= \frac{n}{M} \sum_{j \in B} A_j - I_d, & F_T &= \frac{n}{M} \sum_{j \in T} A_j - I_d, \quad |T| = M. \end{aligned} \quad (10.1)$$

We construct a joint law of a Bernoulli subset B and a uniform M -element subset T . All random choices below take values in finite sets.

For every subset $C \subseteq G$, fix one M -element set $f(C)$ nested with C : choose $f(C) \subseteq C$ when $|C| \geq M$, and choose $C \subseteq f(C)$ when $|C| \leq M$. These choices exist by deleting or adding elements of the finite set G ; when $|C| = M$, necessarily $f(C) = C$. Draw B_0 by independent Bernoulli selectors of mean ρ , and independently draw a uniform permutation π of G . Define

$$(B, T) = (\pi(B_0), \pi(f(B_0))), \quad K = |B| = |B_0|, \quad \begin{cases} B \subseteq T, & K \leq M, \\ T \subseteq B, & K \geq M. \end{cases} \quad (10.2)$$

For each fixed permutation, $\pi(B_0)$ has the same independent Bernoulli law as B_0 , because the probability of a subset depends only on its cardinality. Thus B has the required Bernoulli marginal. For every fixed value of B_0 , the image of the fixed M -element set $f(B_0)$ under a uniform permutation is uniform among all M -element sets: each possible image has exactly $M!(n-M)!$ preimages. In particular, T is uniform even after conditioning on B_0 . Consequently K and T are independent, and for every M -element set t and $0 \leq k \leq n$,

$$\mathbb{P}\{K = k, T = t\} = \frac{1}{\binom{n}{M}} \binom{n}{k} \rho^k (1 - \rho)^{n-k}. \quad (10.3)$$

The joint law of (B, T) is invariant under applying the same permutation to both sets: replacing π by $\sigma\pi$ preserves its uniform law. Given B , permutations fixing B act transitively on its M -element subsets, or on its M -element supersets, as applicable. Together with nestedness, this shows that the same joint law can be described by uniformly thinning B when $K \geq M$, or uniformly filling B from its complement when $K \leq M$.

10.2 Conditional inclusion probabilities and retention

For a real number x , write $x_+ = \max(x, 0)$. The Bernoulli marginal gives $\mathbb{E}K = n\rho = M$. Hence the positive and negative deviations have the same expectation; denote it by $a = \mathbb{E}(K - M)_+ = \mathbb{E}(M - K)_+$.

Fix an M -element set t . Joint permutation invariance implies that $\mathbb{P}\{j \in B \mid T = t\}$ is constant over $j \in t$, and also constant over $j \notin t$. Denote these two values by p_{in} and p_{out} . Nestedness gives $|t \setminus B| = (M - K)_+$ and $|B \setminus t| = (K - M)_+$ on the conditional support. Since the conditional law of K is its unconditional law by (10.3), summing the inclusion probabilities inside and outside t gives

$$M(1 - p_{\text{in}}) = a, \quad (n - M)p_{\text{out}} = a, \quad p_{\text{in}} = 1 - \frac{a}{M}, \quad p_{\text{out}} = \frac{a}{n - M}. \quad (10.4)$$

Put $v = M(n - M)/n$. Using $\sum_j A_j = I_d$ in the conditional expectation of (10.1), the two coefficients in (10.4) yield the exact matrix identity

$$c := p_{\text{in}} - p_{\text{out}} = 1 - \frac{an}{M(n - M)} = 1 - \frac{a}{v}, \quad (10.5)$$

$$\mathbb{E}[E_\rho(B) \mid T = t] = \frac{n}{M}(p_{\text{in}} - p_{\text{out}}) \sum_{j \in t} A_j + \left(\frac{n}{M}p_{\text{out}} - 1 \right) I_d = cF_t.$$

In the last equality, $(n/M)p_{\text{out}} - 1 = -c$. This calculation does not require the matrices A_j to commute.

The variance of K follows by expanding the square of the sum of its centered selectors. Every off-diagonal term has zero expectation by independence, and each diagonal term contributes $\rho(1 - \rho)$. Also, $(M - 1)(n - M - 1) \geq 0$ implies $M(n - M) \geq n - 1$. Since $1 \leq M < n$ forces $n \geq 2$, we obtain

$$v = \mathbb{E}(K - M)^2 = n\rho(1 - \rho) = \frac{M(n - M)}{n} \geq \frac{n - 1}{n} \geq \frac{1}{2}, \quad (10.6)$$

$$4a^2 = (\mathbb{E}|K - M|)^2 \leq v.$$

The last inequality is Cauchy–Schwarz for a probability law; the equality preceding it uses $\mathbb{E}|K - M| = 2a$. As $v \geq 1/2$, squaring both nonnegative sides shows $\sqrt{v}/2 \leq 3v/4$. Therefore

$$0 \leq a \leq \frac{\sqrt{v}}{2} \leq \frac{3v}{4}, \quad \frac{1}{4} \leq c = 1 - \frac{a}{v} \leq 1. \quad (10.7)$$

This estimate includes $M = 1$ and $M = n - 1$; no asymptotic estimate for a binomial tail is used.

10.3 Trace-power convexity by the Hessian argument

The convexity used here is a standard instance of the convex spectral function principle of Davis [14]. The following Hessian calculation supplies the particular finite-dimensional proof needed for the sampling comparison; it does not assert operator convexity of the matrix power. For every integer $q \geq 0$, the function $\Phi_q(A) = \text{tr}(A^{2q})$ is convex on the real vector space of symmetric d -by- d matrices. We give a finite-dimensional second-derivative proof. When $q = 0$, the function is the constant d . Hence suppose $q \geq 1$, and write $p = 2q$.

Fix arbitrary real symmetric matrices A, V , and let $C(t) = A + tV$ and $f(t) = \text{tr}(C(t)^p)$. The matrix entries are polynomials in t . Differentiating the noncommutative product and using cyclicity of trace gives

$$\begin{aligned} f'(t) &= p \text{tr}(VC(t)^{p-1}), \\ f''(t) &= p \sum_{k=0}^{p-2} \text{tr}(VC(t)^{p-2-k}VC(t)^k). \end{aligned}$$

Indeed, the derivative of $C(t)^p$ is the sum of the p products obtained by replacing one factor by V ; their traces all equal $\text{tr}(VC(t)^{p-1})$. Differentiating this expression once more gives the second line.

At any fixed t , diagonalize $C(t)$ as $Q^\top C(t)Q = \text{diag}(\lambda_1, \dots, \lambda_d)$ with Q orthogonal, and set $V' = Q^\top VQ$. Trace is invariant under this conjugation, and V' is symmetric. Expanding each trace in coordinates and collecting the finite sums yields

$$\begin{aligned} f''(t) &= p \sum_{i,j} (V'_{ij})^2 \sum_{k=0}^{p-2} \lambda_i^k \lambda_j^{p-2-k} \\ &= p \sum_{i,j} (V'_{ij})^2 g_{p-1}(\lambda_i, \lambda_j), \\ g_\ell(x, y) &= \sum_{k=0}^{\ell-1} x^k y^{\ell-1-k}. \end{aligned}$$

The diagonalization evaluates the Hessian at this fixed t ; no differentiability of eigenvalues or eigenvectors is required.

For every positive odd integer ℓ , the kernel g_ℓ is nonnegative on $\mathbb{R}^2$. If $x \neq y$, the geometric sum identity gives

$$g_\ell(x, y) = \frac{x^\ell - y^\ell}{x - y} \geq 0,$$

because the odd power $x \mapsto x^\ell$ is increasing on $\mathbb{R}$. If $x = y$, each term in the defining sum is $x^{\ell-1}$, so $g_\ell(x, x) = \ell x^{\ell-1} \geq 0$, since $\ell - 1$ is even. Here $\ell = p - 1$ is positive and odd. Consequently $f''(t) \geq 0$ for every real t .

Thus Φ_q is convex along every symmetric affine line. Taking the line from any symmetric matrix X to any symmetric matrix Y proves convexity on the entire symmetric matrix space. Finite Jensen follows by induction on the number of nonzero weights from this two-point convexity inequality. For any finite family of symmetric matrices A_ω and weights $\theta_\omega \geq 0$ with $\sum_\omega \theta_\omega = 1$,

$$C = \sum_\omega \theta_\omega A_\omega, \quad \text{tr}(C^{2q}) \leq \sum_\omega \theta_\omega \text{tr}(A_\omega^{2q}). \quad (10.8)$$

Apply (10.8) to the finite conditional law of B given $T = t$. Equation (10.5) gives $c^{2q} \text{tr}(F_t^{2q}) \leq \mathbb{E}[\text{tr}(E_\rho(B)^{2q}) \mid T = t]$. Average over the uniform law of T , and use (10.7) and nonnegativity of even trace powers. The resulting comparison is

$$\begin{aligned} c^{2q} \mathbb{E}_T \text{tr}(F_T^{2q}) &\leq \mathbb{E}_B \text{tr}(E_\rho(B)^{2q}), \\ \mathbb{E}_T \text{tr}(F_T^{2q}) &\leq 4^{2q} \mathbb{E}_B \text{tr}(E_\rho(B)^{2q}), \quad \rho = M/n. \end{aligned} \quad (10.9)$$

The expectations on the two sides use the required uniform fixed-size and independent Bernoulli marginals. Thus the comparison concerns the actual sample of distinct rows.

10.4 Averaging the signs and matching limited independence

Return to $W = HD_y HD_x U$, and write $G_M = (n/M) \sum_{j \in T} A_j = I_d + F_T$ for the actual sampled Gram. The coupling is chosen independently of both sign families. Since (10.9) holds for every fixed W , first average it under fully independent unbiased signs, and then apply Theorem 1.3 with Bernoulli mean $\rho n = M$.

For the limited-independence law in Theorem 1.1, the fixed-size moment equals this fully independent fixed-size moment. Indeed, for each fixed row set T , every entry of $G_M - I_d = F_T$ has degree at most two in each sign family. Its even trace power has separate coordinate degrees at most $4q$ and $4q$. Section 8 therefore gives exact expectation matching for independent sign families with the prescribed $\min(n, 4q)$ -coordinate marginals. The row set has the same uniform law on both sides and is independent of the signs, so averaging this equality over T preserves it. Writing $\mathbb{E}_{\text{lim}}$ for this possibly limited-independent sign law and $\mathbb{E}_{\text{iid}}$ for fully independent signs, we have

$$\begin{aligned} \mathbb{E}_{\text{lim}, T} \text{tr}((G_M - I_d)^{2q}) &= \mathbb{E}_{\text{iid}, T} \text{tr}((G_M - I_d)^{2q}) \\ &\leq 4^{2q} \mathbb{E}_{\text{iid}, B} \text{tr}(E_{M/n}^{2q}) \\ &\leq d \left[4 \left(6\sqrt{D_q/M} + 9D_q/M \right) \right]^{2q}, \quad D_q = d + 27q^3. \end{aligned} \tag{10.10}$$

Only the sign moments were matched in this argument. The auxiliary Bernoulli selectors remain fully independent, and the final fixed-size row set retains its full uniform law.

11 Proof of the fixed-size theorem and row order

We now combine the exact fixed-size moment with Markov's inequality. The full-row endpoint is deterministic. In the remaining case, the explicit numerical budget makes the moment root smaller than $\epsilon/2$; the choice of q then gives the desired failure probability.

11.1 Constants and endpoints

Take $q = \lceil \log(4d/\delta)/(2 \log 2) \rceil$, $D_q = d + 27q^3$, and $M = \min(n, \lceil 8192D_q/\epsilon^2 \rceil)$, as in Theorem 1.1. Under $d \geq 1$, $0 < \epsilon < 1$, and $0 < \delta < 1/2$, both $q \geq 1$ and $M \geq 1$.

If $M = n$, the sampled Gram is $W^\top W = I_d$ for every realization of the signs, by (1.4). Its failure probability is zero. This includes $n = 1$. In the remaining case $1 \leq M < n$, the minimum defining M implies $M \geq 8192D_q/\epsilon^2$, and hence

$$\begin{aligned} \frac{D_q}{M} &\leq \frac{\epsilon^2}{8192}, \quad \sqrt{\frac{D_q}{M}} \leq \frac{\epsilon}{64}, \\ 4 \left(6\sqrt{D_q/M} + 9D_q/M \right) &\leq \frac{3}{8}\epsilon + \frac{9}{2048}\epsilon^2 \leq \frac{777}{2048}\epsilon < \frac{\epsilon}{2}. \end{aligned} \tag{11.1}$$

The square-root estimate uses $8192 \geq 64^2$, and the last two inequalities use $0 < \epsilon < 1$ and $777 < 1024$.

For every real symmetric matrix A , $\|A\|^{2q} \leq \text{tr}(A^{2q})$, since the trace is the sum of the nonnegative even powers of its eigenvalues. Markov's inequality, (10.10), and (11.1) now give

the following bound under the actual limited-independent sign and uniform-row product law:

$$\begin{aligned}\mathbb{P}_{\text{lim},T}\{\|G_M - I_d\| \geq \epsilon\} &\leq \frac{\mathbb{E}_{\text{lim},T} \text{tr}((G_M - I_d)^{2q})}{\epsilon^{2q}} \\ &\leq d 2^{-2q} = d 4^{-q} \leq \frac{\delta}{4} \leq \delta.\end{aligned}\tag{11.2}$$

Indeed, the definition of q gives $4^q \geq 4d/\delta$. Since the event in (1.8) is contained in the event in (11.2), this proves Theorem 1.1, with the stronger bound $\delta/4$ in the proper-sampling case. Whenever $\|G_M - I_d\| \leq \epsilon$, the spectral theorem gives $(1 - \epsilon)I_d \preceq G_M \preceq (1 + \epsilon)I_d$, equivalently the claimed two-sided squared-norm inequalities on the input subspace. The independence and uniformity of the sampled rows are exactly those established by the finite construction above.

11.2 Removing the apparent logarithmic dependence on dimension

Write $h = \log(1/\delta)$, $a = \lceil \log_4 d \rceil$, and $b = \lceil \log_4(4/\delta) \rceil$. These quantities are nonnegative. Subadditivity of the ceiling gives

$$q = \lceil \log_4 d + \log_4(4/\delta) \rceil \leq a + b.\tag{11.3}$$

For every nonnegative integer j , $j^3 \leq 4^j$. Check $j = 0, 1, 2$; for $j \geq 2$, induction uses $(j+1)^3 \leq (3j/2)^3 \leq 4j^3$. Since $a < \log_4 d + 1$, this yields $a^3 \leq 4^a < 4d$. Also $\log 2 = \int_1^2 x^{-1} dx \geq 1/2$, so $h \geq 1/2$ and $\log 4 \geq 1$. Therefore $b < 2 + h/\log 4 \leq 2 + h \leq 5h$.

For nonnegative a, b , the identity $4(a^3 + b^3) - (a + b)^3 = 3(a - b)^2(a + b) \geq 0$ gives

$$\begin{aligned}q^3 &\leq 16d + 500 \log^3(1/\delta), \\ D_q &= d + 27q^3 \leq 13500[d + \log^3(1/\delta)].\end{aligned}\tag{11.4}$$

The ceiling in the definition of M adds less than one. Since $d \geq 1$ and $0 < \epsilon < 1$, that additive one is at most $[d + \log^3(1/\delta)]/\epsilon^2$. Hence

$$\begin{aligned}M &\leq \min \left\{ n, \frac{110592001}{\epsilon^2} [d + \log^3(1/\delta)] \right\}, \\ M &= O\left(\epsilon^{-2} [d + \log^3(1/\delta)]\right).\end{aligned}\tag{11.5}$$

This proves Corollary 1.2. For fixed δ , it is $O(d/\epsilon^2)$ with a constant independent of n, d . The large numerical coefficient records a conservative proof bound rather than a practical tuning recommendation.

12 Scope and limitations

The theorem concerns real Walsh matrices with $n = 2^k$, an arbitrary fixed orthonormal input frame, and a uniform set of distinct sampled rows. It establishes the constant-failure dimension order in Problem 5.6 of Amsel et al. [1] for this model. There is no additional relation between n and d beyond $1 \leq d \leq n$. The constants are explicit proof bounds; their numerical optimization is not part of the result.

The probabilistic and operator estimates used by this proof are derived here. This is a statement about proof dependencies, not originality of Walsh analysis, low-degree moment matching, Schur bounds, or convex sampling comparisons. The overlapping fixed-failure theorem of Yang is acknowledged above [5]. Character orthogonality gives the row functional; finite-product multiplication gives the row channels; their collective estimate gives small-subset control. Positivity then bounds the selector operator. Exact vacuum moments, polynomial matching,

and a finite coupling complete the passage to a fixed-size embedding. In particular, the argument uses the actual sample without replacement throughout that final passage.

Extensions to complex Fourier matrices or other trigonometric transforms require separate analysis. The limited-independence result applies to the sign families; the final row sample still has its full uniform law. No low-randomness construction of that fixed-size sample is asserted. The confidence obstruction in Appendix A rules out a uniform Gaussian confidence term for finitely many Walsh/sign rounds, but does not show that the cubic logarithm in our upper bound is optimal.

The finite-product and polynomial-matching methods were motivated by the SparseStack development [2] and associated research notes. The required identities and inequalities are derived in this paper. Appendix B records the formal artifact, the exact verification scope, and the distinction between mathematical proofs and finite computational checks.

A Confidence obstruction for finitely many rounds

A.1 Alternating sparse supports

This appendix is independent of the proof above. Fix a positive integer number of rounds t , fully independent sign matrices $D_1, \dots, D_t$, and a uniform sample of M distinct rows. Let $L \geq 2$ be a power of two, $n = L^2$, and $H = H_L \otimes H_L$. Define

$$u = \frac{\mathbf{1}_L}{\sqrt{L}} \otimes e_0, \quad v = e_0 \otimes \frac{\mathbf{1}_L}{\sqrt{L}}. \quad (\text{A.1})$$

Both are unit vectors with L nonzero coordinates. By character orthogonality, $Hu = v$ and $Hv = u$. In round j , require all signs to be $+1$ on the deterministic current support: the support of u at odd rounds and of v at even rounds. The events use different independent sign layers, each has probability 2^{-L} , and on their intersection the state alternates exactly between u and v .

Thus, for $S_t = \sqrt{n/M} RHD_t \cdots HD_1$, and $1 \leq M \leq n - L$,

$$\mathbb{P}\{\|S_t u\|^2 < 1 - \epsilon\} \geq 2^{-tL} \frac{\binom{n-L}{M}}{\binom{n}{M}}, \quad 0 < \epsilon < 1. \quad (\text{A.2})$$

The binomial ratio is the probability that the row sample misses the final L -point support. On that event the sketched vector is exactly zero.

A.2 A finite necessary row count

The miss probability can also be written by choosing which support points avoid the sample:

$$\frac{\binom{n-L}{M}}{\binom{n}{M}} = \frac{\binom{n-M}{L}}{\binom{n}{L}} = \prod_{j=0}^{L-1} \left(1 - \frac{M}{n-j}\right) \geq \left(1 - \frac{M}{n-L+1}\right)^L. \quad (\text{A.3})$$

For $M \leq n - L$, all displayed factors are positive, and $n - j \geq n - L + 1$ proves the inequality. Any guarantee of failure at most $\delta < 2^{-tL}$ therefore requires

$$M \geq (n - L + 1) \left(1 - 2^t \delta^{1/L}\right). \quad (\text{A.4})$$

For $M \geq n - L + 1$, (A.4) is automatic, so the necessary inequality holds for all M . For two rounds and $\delta = 2^{-4L}$, it becomes $M \geq \frac{3}{4}(n - L + 1)$.

A.3 Incompatibility with a uniform linear-logarithmic confidence term

Fix $a > t \log 2$, and let $\delta_L = e^{-aL}$. A proposed guarantee of $M \leq C\epsilon^{-2}(1 + \log(1/\delta_L))$, with fixed C, ϵ, t , uses $M = O(L)$. But (A.4) requires a positive constant fraction of L^2 , because $1 - 2^t e^{-a} > 0$. These two conclusions are incompatible for large L .

Therefore the uniform Gaussian-style confidence term $d + \log(1/\delta)$ cannot hold for every confidence level for any fixed number of these Walsh/sign rounds. This does not contradict Theorem 1.1: when $\log(1/\delta)$ is of order $L = \sqrt{n}$, its cubic confidence term reaches the full-row endpoint.

A.4 A stronger two-round alignment event

On L sites, the $2L$ signed Walsh characters are distinct sign patterns, each of probability 2^{-L} . Requiring the first diagonal on the support of u to equal a signed character produces $\pm e_a \otimes \mathbf{1}_L / \sqrt{L}$ after the first Walsh transform. Given that outcome, the fresh second diagonal on its support can again be any signed character. After the next transform, the vector is $\pm(H_L e_a) \otimes e_b$, still supported on L coordinates.

The total probability of this disjoint alignment family is $(2L/2^L)^2 = 4L^2 2^{-2L}$. Therefore the factor 2^{-2L} in the two-round version of (A.2) can be strengthened to $4L^2 2^{-2L}$. This is a sufficient family of bad events, not a classification of all failures.

B Formalization and verification

The accompanying development uses Lean 4.33.0 and Mathlib at [commit db584cd6](#). The source and evidence [3] are available in the [project repository](#). Here, formal verification means that the stated formal propositions have complete proof terms accepted by the specified kernels. It does not mean that a proof assistant has checked the manuscript’s prose or its interpretation of an external open problem.

Verified snapshot. On 9 September 2026, [verification run 34373116757](#) passed for [source commit f813b5c](#). The run completed a clean Linux build and 33 transitive axiom audits. The audited declarations depend only on `propext`, `Classical.choice`, and `Quot.sound`; the completed Solution and proof library contain no unfinished proofs or additional mathematical axioms. The count consists of the 32 underlying declarations audited in the earlier 7 September development and the additional Challenge/Solution theorem. The [archived evidence](#) records the checked source hashes, tool revisions, build output, and audit results. Its later archival commit does not change the checked proof sources.

Mathematical coverage. The fixed-size embedding theorem and its norm-failure form are `SRHT.two_round_srht_ose` and `SRHT.two_round_srht_norm_failure`. Their assumptions allow each sign family to match independent unbiased signs on at most $\min(n, 4q)$ coordinates; the two families and the uniform distinct-row sample have the stated product law. The arbitrary-rate Bernoulli moment theorem and its limited-independence form are `SRHT.BernoulliGeneral.moment_bound` and `SRHT.BernoulliGeneral.moment_bound_limited`. The latter requires the separate matching orders $\min(n, 4q)$, $\min(n, 4q)$, and $\min(n, 2q)$ for the two sign families and the selectors. The combined embedding and cubic-logarithmic row bound is `SRHT.two_round_srht_ose_log_confidence`. The support-alignment construction, exact miss probability, necessary row bound, stronger two-round event, and incompatibility with a universal Gaussian confidence budget are proved in the `Alignment`, `SupportMiss`, `AlignmentObstruction`, `StrongAlignment`, and `GaussianObstruction` modules. The [coverage table](#) maps the mathematical results and principal operator and sampling lemmas to

their formal declarations. The exposition follows these arguments, without reproducing the proof-assistant syntax line by line.

Statement comparison and independent replay. The same Linux run passed Comparator for the single selected theorem `RerandomizedSTT.problem_5_6`. Comparator matched the separately compiled Challenge and Solution statements and their definition dependencies; NanoDa and Lean’s default kernel both accepted the exported solution proof. This selected theorem uses fully independent uniform signs, the literal finite uniform law on distinct-row samples, and the prescribed row count M . It proves the simultaneous squared-norm embedding event with success probability at least $1 - \delta$, together with

$$1 \leq M \leq n, \quad M \leq \lceil 8192(1 + 432/\delta) + 1 \rceil d/\epsilon^2.$$

Thus the compared statement gives the requested $O(d/\epsilon^2)$ bound at every fixed failure probability. The other results listed above were built and audited in Lean; they were not separately selected for Comparator or NanoDa replay. The Challenge contains one intentional theorem placeholder for comparison. It is not imported by the completed Solution or its audit.

Scope and presentation status. The model is real Walsh space with $n = 2^k$, and the input subspace is fixed before sampling. The result concerns neither all trigonometric transforms nor a simultaneous embedding of every subspace by one draw. Lean does not certify publication priority, attribution, the bit-level implementation of a sampler, or the manuscript–source correspondence. The recorded work has not been submitted to or registered by Palomar, and no independent human peer review is claimed.

As of 9 September 2026, a separate [Challenge-rendering diagnostic](#) records a failure in Palomar’s core-notation audit on a dependency involving `ZMod 2`. The ordinary Lean and Verso literate builds pass. A compatibility change passes the notation audit on a copied Challenge, but has not been applied to the checked Challenge or Solution. The repository therefore has the successful proof checks described above, while the full Palomar rendering pipeline remains blocked. These are separate checks.

Reproduction and manuscript provenance. The pinned repository contains `lean-toolchain`, `lake-manifest.json`, `Audit.lean`, and the verification scripts. A fresh checkout of the checked source revision can reproduce the Lean build and axiom inventory with

```
lake exe cache get
lake build
lake env lean Audit.lean
```

Comparator and independent replay additionally use the pinned Linux workflow and its confinement tools. The archived results identify the commands and tool revisions actually used; a new build is a new verification run.

This manuscript revises the exposition of the project’s 7 September 2026 technical report. The current editable source is `SRHT.tex`; its typeset companion is `SRHT.pdf`. The original report and the [source-provenance record](#) remain part of the artifact. The historical 26 numerical and symbolic checks concern finite instances of the operator identities and sampling model. They support the development but do not replace its general proofs. The examples and expanded proof guide in this revision are explanatory; the kernel checks apply to the formal declarations identified above.

C Notation index

Symbol	Meaning
$n = 2^k, G$	Ambient dimension and the coordinate group $\mathbb{F}_2^k$.
d, U	Subspace dimension and an orthonormal input frame.
u_i, P, ℓ_i	Input row vector, projection $UU^\top$, and leverage score $\ u_i\ ^2$.
H, χ_b, f_b	Normalized Walsh matrix, Walsh character, and normalized character vector.
x, y, D_x, D_y	Two sign families and their diagonal matrices.
W, w_j	Rerandomized frame HD_yHD_xU and its row vectors.
M, R, S	Fixed sample size, row selector, and normalized sketch.
ρ, m, E_ρ	Bernoulli rate, expected sample size, and centered Gram error.
q, D_q	Half the even moment order and $d + 27q^3$.
$\mathcal{X}_r, \mathcal{Y}_s$	Occupation spaces for the two sign species.
$a_i, a_i^\dagger, N_i$	Annihilation, creation, and occupancy at one sign site.
$c_j, c_j^\dagger$	Selector annihilation and creation.
$P_{\text{sig}}, P_{\text{sel}}$	Sign and selector degree cutoffs.
A_j	Positive row effect after sign compression.
Ω	Constant function, or vacuum vector, in the product space.

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